



CELEBAL

TECHNOLOGIES

E-COMMERCE ANALYTICS SYSTEM USING
PYTHON, SQLITE & SQL

Celebal Summer Internship 2026

Week 8 Assignment

End-to-End E-Commerce Data Processing, SQL Analytics &
Business Reporting System

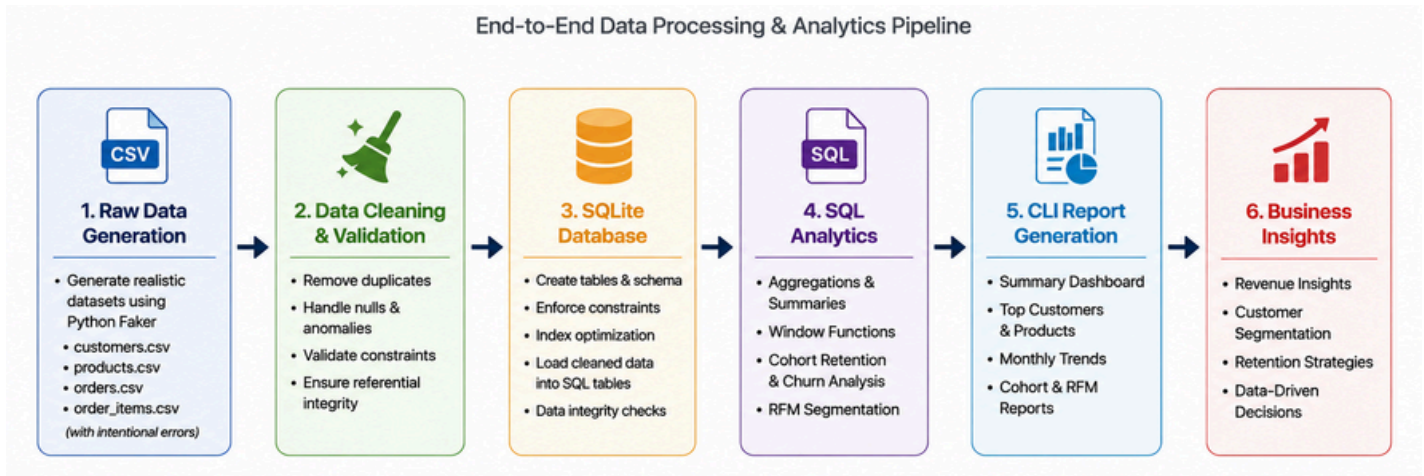
**Dataset: Synthetic E-Commerce Dataset (Customers, Products,
Orders & Order Items)**

Submission Date:
12-07-2026

Prepared By:
Keshav Kumar Sharma

1. Project Overview & Architecture

This project demonstrates a complete data analytics pipeline for an E-Commerce platform. It covers data generation, data cleaning, database design, SQL-based analytics, automated testing, and report generation. The solution transforms raw transactional data into meaningful business insights using Python, Pandas, SQLite, and SQL.



Core Pipeline Steps

Step 1: Raw Data Generation

Creates mock customers, products, orders, and items using Faker. Inserts intentional data duplicates, negative values, and future order dates to simulate real-world data issues.

Step 2: Sanitization & Cleaning

Uses Pandas to parse dates, drop duplicate IDs, clean emails via Regex, filter out negative values, and drop orphan items to secure referential integrity.

Step 3: Relational DB Load

Imports sanitized datasets into structured SQLite tables. Enables constraints (check constraint, NOT NULL, FOREIGN KEYS) and establishes indexing on frequently joined columns.

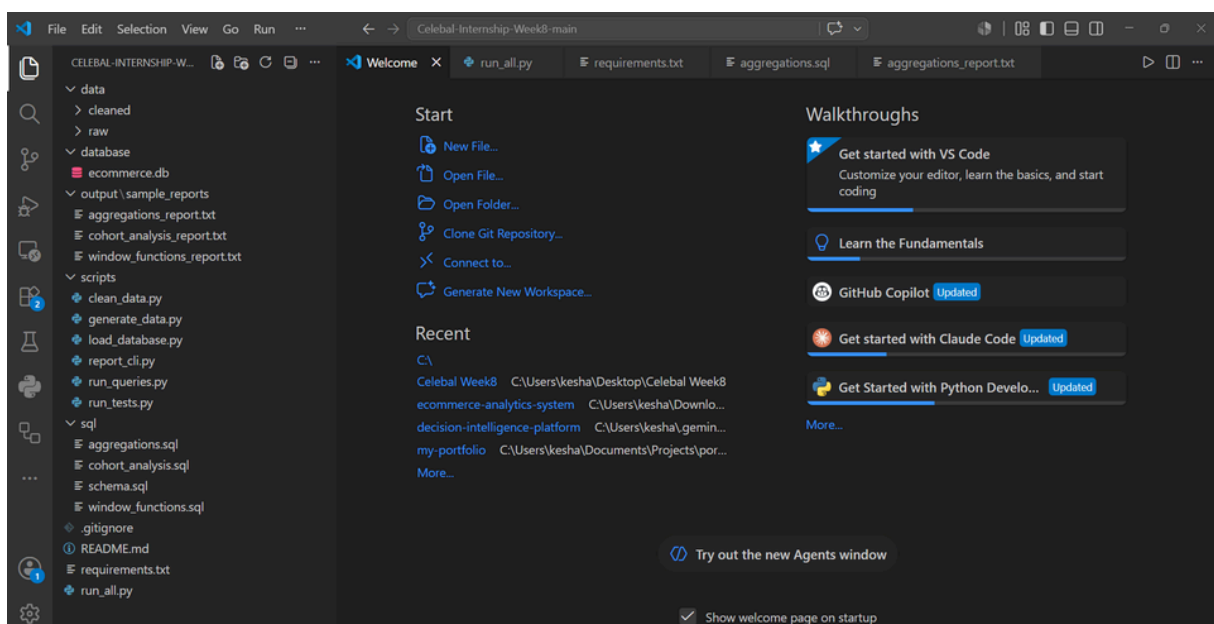


Fig : Folder structure

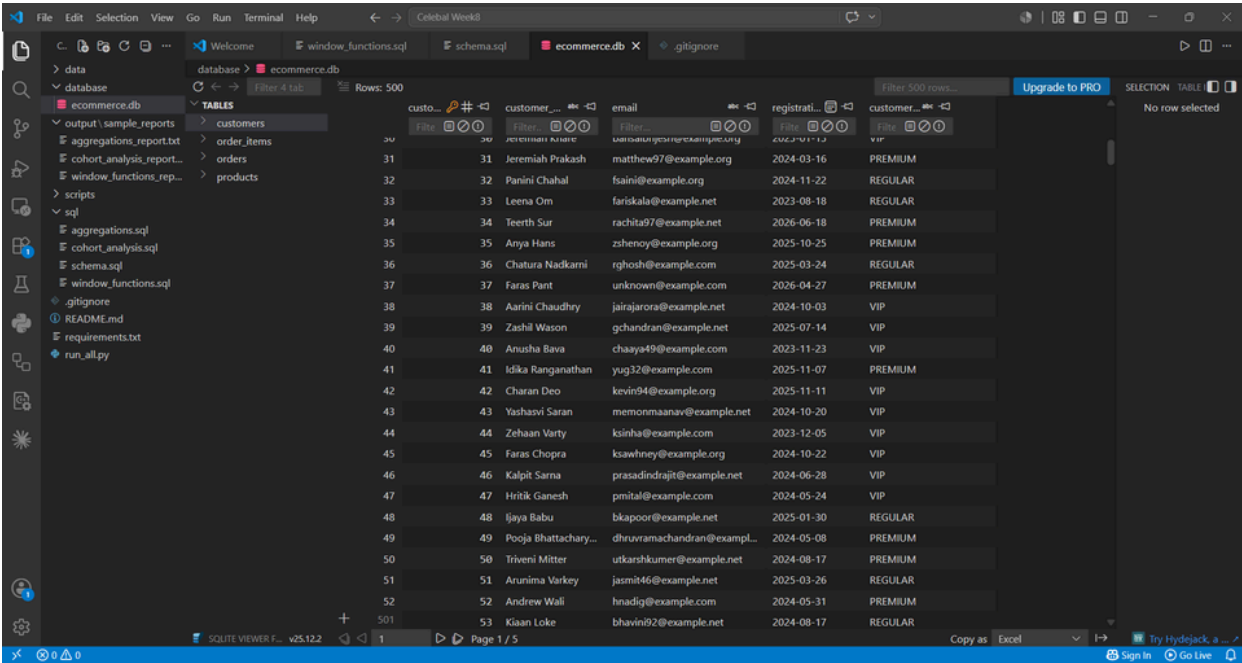
2. Database Schema & Volume Statistics

Loaded Database Row Counts

Table Name	Description	Record Count
customers	Customer Profile Details	500
products	Product catalog & Categories	495
orders	Order dates & status tracking	490
order_items	Sales line items & margins	1420

Database Table Schemas

CUSTOMERS TABLE	customer_id (PK) , customer_name , email , customer_type
PRODUCTS TABLE	product_id (PK), product_name, category, price
ORDERS TABLE	order_id (PK), customer_id (FK), order_date, status
ORDER_ITEMS TABLE	item_id (PK), order_id (FK) , product_id (FK) , quantity , unit_price , discount_percent



Index Optimization

The cleaned datasets are loaded into SQLite after validation. Primary keys, foreign keys, and indexes are created to ensure referential integrity and improve query execution performance.

3. Data Generation & Data Cleaning

Dataset Generation

The project begins by generating realistic e-commerce datasets using the Python Faker library along with the random module. Four independent CSV files are created to simulate a real-world online retail environment. Intentional anomalies such as duplicate records, invalid values, missing data, and future dates are introduced to test the robustness of the data cleaning pipeline.

Generated Datasets

Dataset	Description
customers.csv	Customer profile information
products.csv	Product catalog and pricing
orders.csv	Customer order transactions
order_items.csv	Individual products purchased in each order

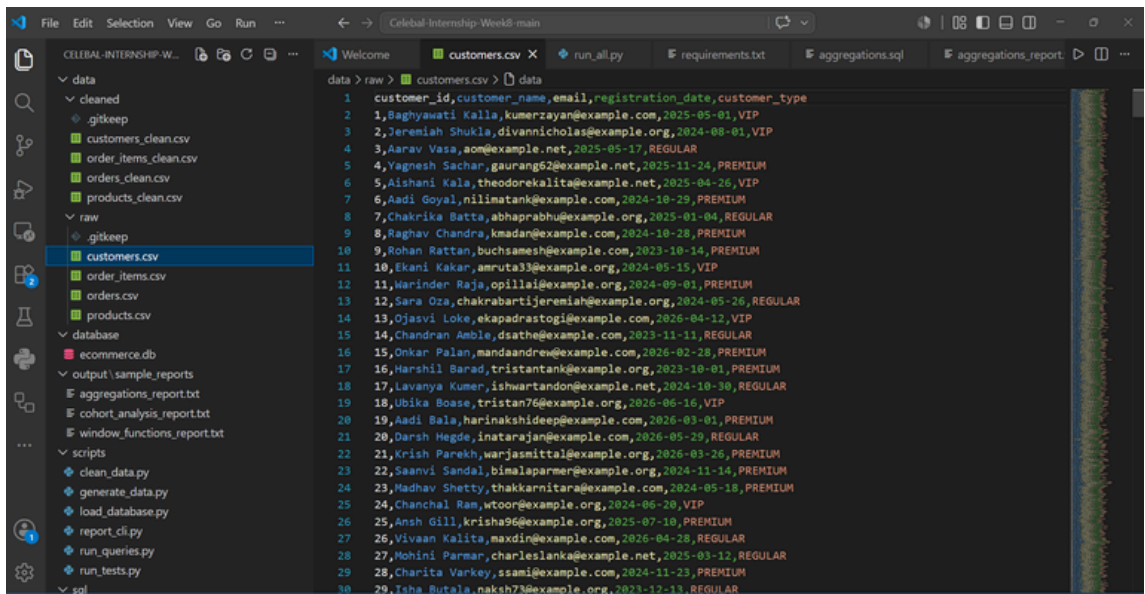


Fig : Generated data

Data Cleaning & Validation

After generating the raw datasets, a comprehensive data cleaning process is performed using Pandas. The pipeline validates data quality, removes inconsistencies, and prepares the datasets for database loading.

```

1 import os
2 import pandas as pd
3
4 RAW_PATH = "data/raw"
5 CLEAN_PATH = "data/cleaned"
6
7 os.makedirs(CLEAN_PATH, exist_ok=True)
8
9 def clean_orders(orders):
10
11     orders = orders.copy()
12
13     orders["customer_id"] = (

```

PROBLEMS OUTPUT TERMINAL

```

Negative Prices Removed : 5
Negative Quantity Removed : 10
Invalid Discounts Removed : 5
Invalid Unit Prices Removed : 5
Invalid Order IDs Removed : 46
Invalid Product IDs Removed : 8

Final Dataset Sizes
Customers : 500
Products : 495
Orders : 490
Order Items : 1426

Cleaned CSV Files Saved Successfully!

```

Fig : Cleaned_data.py

4. SQLite Database Implementation

Objective

To create a structured SQLite relational database by importing the cleaned datasets and establishing relationships between tables using primary and foreign keys.

Tasks Performed

- Created SQLite database (ecommerce.db)
- Created relational database tables
- Applied Primary Key constraints
- Established Foreign Key relationships
- Loaded cleaned CSV datasets into SQLite
- Verified row counts after successful data loading

database > ecommerce.db

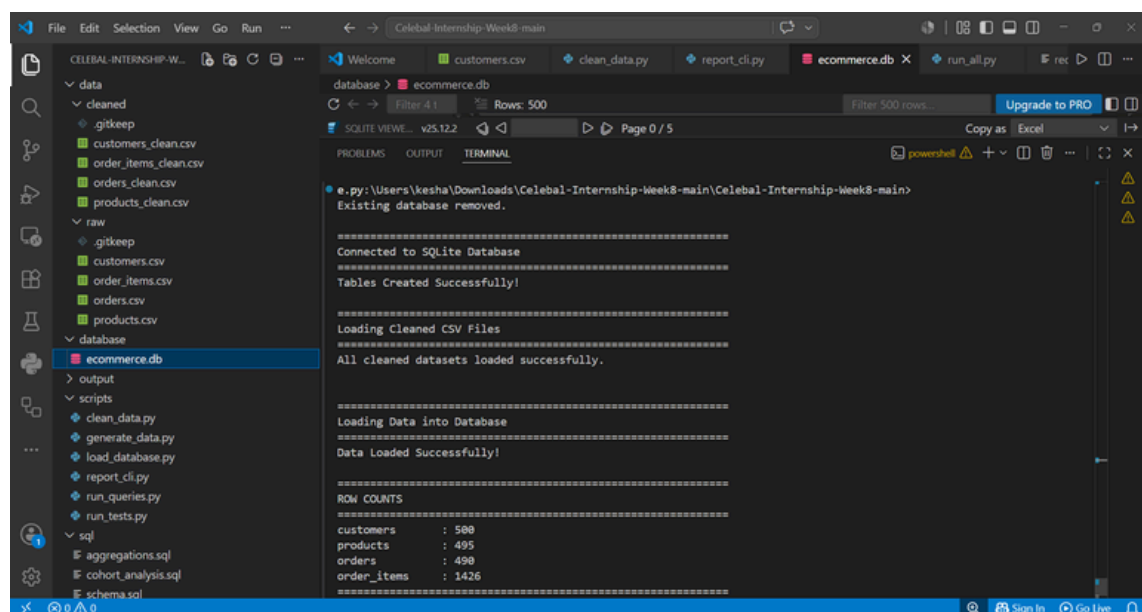
Filter 4.1 Rows: 500

TABLES

- customers
- order_items
- orders
- products

	customer_id	customer_name	email	registration_date	customer_type
1	1	Baghyawati Kalla	kumerzayan@example.com	2025-05-01	VIP
2	2	Jeremiah Shukla	divannicholas@example.org	2024-08-01	VIP
3	3	Aarav Vasa	aom@example.net	2025-05-17	REGULAR
4	4	Yagnesh Sachar	gaurang62@example.net	2025-11-24	PREMIUM
5	5	Aishani Kala	theodorekalita@example.com	2025-04-26	VIP
6	6	Aadi Goyal	nilimatank@example.com	2024-10-29	PREMIUM
7	7	Chakrika Batta	abhprabhu@example.org	2025-01-04	REGULAR
8	8	Raghav Chandra	kmadan@example.com	2024-10-28	PREMIUM
9	9	Rohan Rattan	buchsameash@example.com	2023-10-14	PREMIUM
10	10	Ekani Kakar	amruta33@example.org	2024-05-15	VIP
11	11	Warinder Raja	opillai@example.org	2024-09-01	PREMIUM
12	12	Sara Oza	chakrabartijeremiah@example.com	2024-05-26	REGULAR
13	13	Ojasvi Loke	ekapadrastogi@example.com	2026-04-12	VIP
14	14	Chandran Amble	dsathe@example.com	2023-11-11	REGULAR
15	15	Onkar Palan	mandaandrew@example.com	2026-02-28	PREMIUM

Fig : Database.db



SQLite Database Successfully Created and Populated with Cleaned Data

5. SQL Query Execution

Objective

To execute multiple SQL scripts on the SQLite database and generate analytical reports containing business insights such as revenue, customer behavior, product performance, and sales trends.

SQL Scripts Executed

SQL Script

aggregations.sql

Purpose

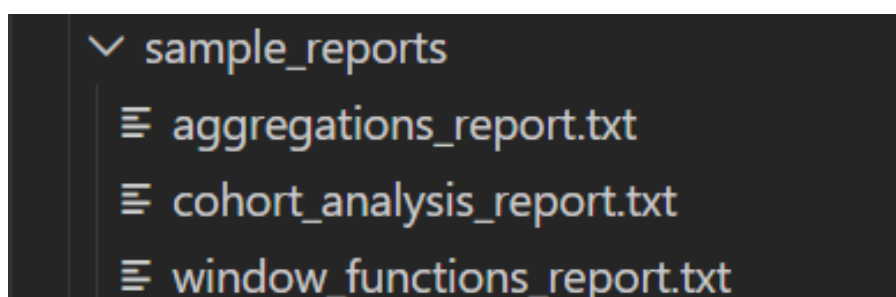
Calculates revenue, top customers, top products, and monthly sales trends

window_functions.sql

Performs ranking, running totals, LAG/LEAD, and window function analysis

cohort_analysis.sql

Performs cohort analysis, RFM segmentation, retention analysis, and customer churn identification



```
PS C:\Users\kesha\Downloads\Celebal-Internship-Week8-main\Celebal-Internship-Week8-main> python scripts/run_queries.py
|      68 | 2023-01-15 |      1275 | CHURNED |
+-----+-----+-----+-----+
|     459 | 2023-01-15 |      1275 | CHURNED |
+-----+-----+-----+-----+
|     266 | 2023-02-01 |      1258 | CHURNED |
+-----+-----+-----+-----+
|     492 | 2023-02-06 |      1253 | CHURNED |
+-----+-----+-----+-----+
|      60 | 2023-02-11 |      1248 | CHURNED |
+-----+-----+-----+-----+
|     228 | 2023-02-19 |      1240 | CHURNED |
+-----+-----+-----+-----+
|     380 | 2023-02-20 |      1239 | CHURNED |
+-----+-----+-----+-----+
|     208 | 2023-03-31 |      1200 | CHURNED |
+-----+-----+-----+-----+

Showing first 10 of 20 rows...

Report Saved : output/sample_reports/cohort_analysis_report.txt

=====
ALL SQL FILES EXECUTED SUCCESSFULLY
=====
PS C:\Users\kesha\Downloads\Celebal-Internship-Week8-main\Celebal-Internship-Week8-main>
```

Successful Execution of SQL Analytics Scripts using run_queries.py

6. SQL Analytics & Report Generation

SQL Analytics

After successfully loading the cleaned datasets into the SQLite database, multiple SQL analytical queries were executed to transform transactional data into meaningful business insights. The project uses advanced SQL techniques including aggregations, window functions, common table expressions (CTEs), and cohort analysis to evaluate sales performance and customer behavior.

Analytics Performed

Revenue Analysis

- Calculated total system revenue.
- Identified top revenue-generating products.
- Ranked high-value customers based on purchase amount.

Aggregation Queries

- Revenue by product category.
- Monthly revenue trends.
- Average order value.
- Top selling products.

Window Functions

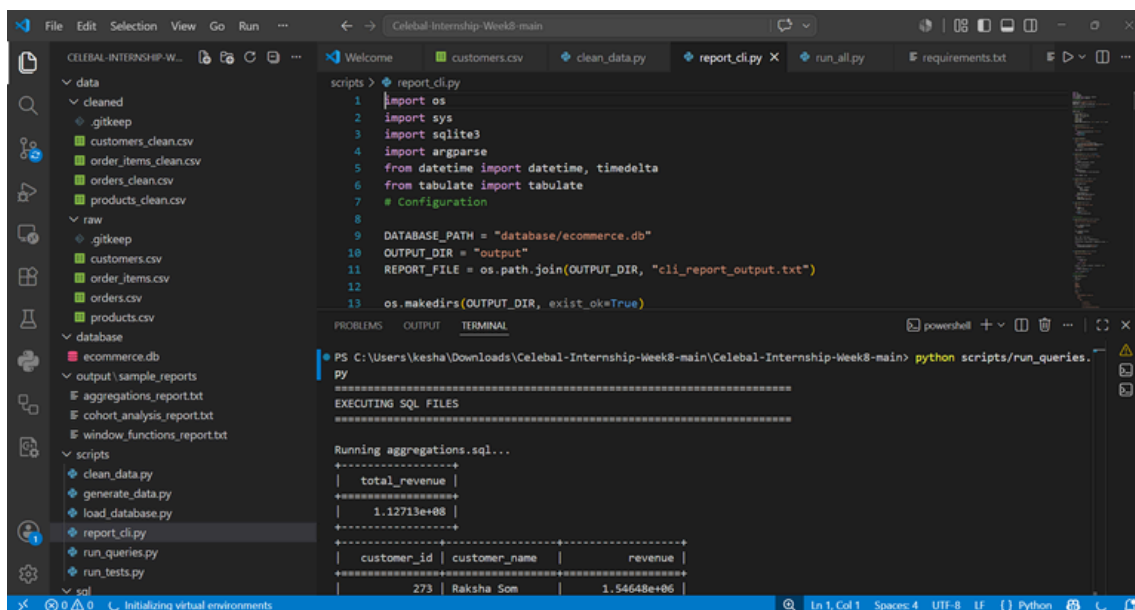
- Customer ranking using RANK() and DENSE_RANK().
- Running monthly revenue totals.
- Previous and next order analysis using LAG() and LEAD().
- Product ranking within each category

Customer Cohort & Segmentation

- First purchase cohort identification.
- Customer retention analysis.
- RFM (Recency, Frequency, Monetary) analysis.
- Spend tier classification (LOW, MEDIUM, HIGH).
- Churned customer identification.

Business Insights Generated

- Total revenue generated by the system.
- Top-performing products by sales.
- High-value customers based on expenditure.
- Monthly revenue growth trends.
- Customer retention and churn analysis.
- Purchase frequency and spending behavior.



The screenshot shows a VS Code editor with a project named 'Celebal-Internship-Week8-main'. The left sidebar displays a file explorer with folders like 'data', 'cleaned', 'raw', 'database', 'output', and 'scripts'. The main editor area shows a Python script 'report_cli.py' with the following code:

```
1 import os
2 import sys
3 import sqlite3
4 import argparse
5 from datetime import datetime, timedelta
6 from tabulate import tabulate
7 # Configuration
8
9 DATABASE_PATH = "database/ecommerce.db"
10 OUTPUT_DIR = "output"
11 REPORT_FILE = os.path.join(OUTPUT_DIR, "cli_report_output.txt")
12
13 os.makedirs(OUTPUT_DIR, exist_ok=True)
```

Below the script, the 'TERMINAL' tab shows the output of running 'python scripts/run_queries.py'. It indicates that SQL files are being executed, specifically 'aggregations.sql'. The output shows the total revenue and a list of top customers:

```
Running aggregations.sql...
+-----+
| total_revenue |
+-----+
| 1.12713e+08 |
+-----+

+-----+
| customer_id | customer_name | revenue |
+-----+
| 273 | Raksha Som | 1.54648e+06 |
+-----+
```

SQL Queries Executed Successfully using run_queries.py

```

PROBLEMS OUTPUT TERMINAL
PS C:\Users\kesha\Downloads\Celebal-Internship-Week8-main\Celebal-Internship-Week8-main> python scripts/run_queries.py
PS C:\Users\kesha\Downloads\Celebal-Internship-Week8-main\Celebal-Internship-Week8-main> python scripts/report_cli.py

=====
E-Commerce Analytics Reporting Tool
=====

Select Trend Type (daily / weekly / monthly): monthly
Enter Start Date (YYYY-MM-DD): 2025-01-01
Enter End Date (YYYY-MM-DD): 2025-06-01

=====
SUMMARY DASHBOARD
=====
+-----+-----+-----+-----+
| Metric | Current | Previous | % Change |
+-----+-----+-----+-----+
| Orders | 83      | 82      | +1.22%   |
+-----+-----+-----+-----+
| Revenue | 2.16032e+07 | 2.1727e+07 | -0.57%   |
+-----+-----+-----+-----+
| Customers | 75      | 79      | -5.06%   |
+-----+-----+-----+-----+
=====

```

Fig : CLI-Based Business Analytics Report

TOTAL SYSTEM REVENUE: Rs. 112,713,000

Top 5 Products by Revenue Contribution

Product Name	Revenue Earned
Dsa Algorithms Model 927	Rs. 1,050,600
Dsa Algorithms Model 337	Rs. 799,148
Hometown Sofa Model 309	Rs. 760,690
Spykar Jeans Model 531	Rs. 726,024
Adidas T-Shirt Model 879	Rs. 710,266

Top 5 High-Value Customers

Customer Name	Revenue
Raksha Som	Rs. 1,546,480
Lopa Kothari	Rs. 1,212,370
Aarini Grewal	Rs. 1,180,670
Ikbal Bora	Rs. 1,175,160
Falak Majumdar	Rs. 1,118,660

Monthly Revenue Trend (Sample Period)

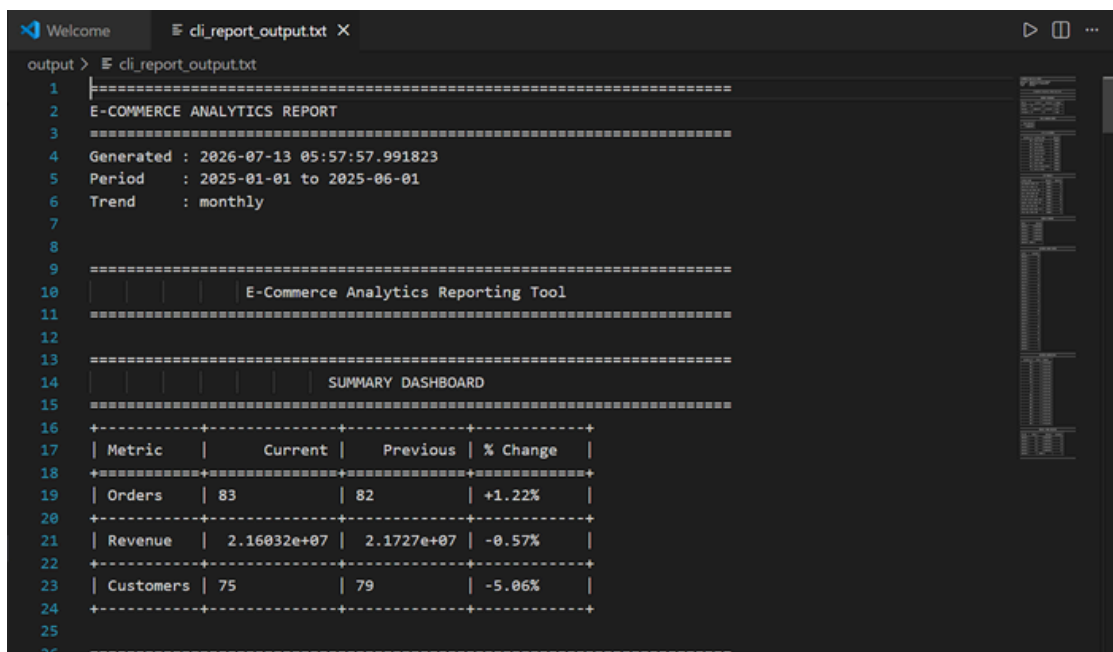
Month	Revenue Earned
2023-01	Rs. 3,196,380
2023-02	Rs. 3,620,520
2023-03	Rs.3,390,220
2023-04	Rs. 5,501,360
2023-05	Rs. 3,626,620
2023-06	Rs. 5,563,990

The analytical reports generated from the SQL queries provide meaningful insights into customer behavior, product performance, sales trends, and revenue distribution. These insights can support data-driven business decisions and strategic planning.

8. CLI Business Report

SQL Analytics

The project includes an interactive Command Line Interface (CLI) reporting tool that retrieves analytical information directly from the SQLite database. The tool allows users to generate business reports by selecting a date range and trend type, making the reporting process flexible and user-friendly.



```
Welcome  cli_report_output.txt X
output > cli_report_output.txt
1 |=====
2 | E-COMMERCE ANALYTICS REPORT
3 |=====
4 | Generated : 2026-07-13 05:57:57.991823
5 | Period    : 2025-01-01 to 2025-06-01
6 | Trend     : monthly
7 |
8 |
9 |=====
10 | | | | | E-Commerce Analytics Reporting Tool
11 |=====
12 |
13 |=====
14 | | | | | SUMMARY DASHBOARD
15 |=====
16 | +-----+-----+-----+-----+
17 | | Metric | | Current | | Previous | | % Change | |
18 | +-----+-----+-----+-----+
19 | | Orders  | | 83      | | 82      | | +1.22%  | |
20 | +-----+-----+-----+-----+
21 | | Revenue | | 2.16032e+07 | | 2.1727e+07 | | -0.57%  | |
22 | +-----+-----+-----+-----+
23 | | Customers | | 75      | | 79      | | -5.06%  | |
24 | +-----+-----+-----+-----+
25 |
26 |=====
```

Fig : Cli Report output

Generated Reports

The CLI reporting tool automatically generates the following business reports:

- Summary Dashboard
- Total Revenue Report
- Top Customers
- Top Products
- Monthly Revenue Trends
- Customer Cohort Report
- Customer Segmentation
- Trend Analysis
- Exported Text Report

7. Testing & Validation

Objective

To verify the correctness, reliability, and data integrity of the complete E-Commerce Analytics pipeline through automated test cases. The validation process ensures that the cleaned datasets satisfy all business rules before analysis and reporting.

Test Results

The automated validation script successfully verified all critical business rules. All five test cases passed without any errors, confirming that the cleaned datasets are consistent and ready for SQL analytics and business reporting.

Verification Check	Logic Evaluated	Status
Orphaned Order Items	Removes line items referencing non-existent orders	PASSED
Discount Validation	Ensures discount percents are bounded between 0% and 100%	PASSED
Zero/Negative Quantity	Discards order lines containing quantity values <= 0	PASSED
Future Order Dates	Filters out any orders placed ahead of current system timestamp	PASSED
Missing Customer Validation	Drops orders lacking a valid associated customer_id	PASSED

```

All test cases passed successfully.
PS C:\Users\keshu\Downloads\Celebal-Internship-Week8-main\Celebal-Internship-Week8-main> python scripts/run_tests.py
=====
Loading Raw Datasets...
=====
Datasets Loaded Successfully!

Cleaning Customers...
Cleaning Products...
Cleaning Orders...
Cleaning Order Items...
Validating Relationships...

=====
DATA CLEANING SUMMARY
=====
Customer Duplicates Removed : 5
Product Duplicates Removed : 5
Order Duplicates Removed : 0
Item Duplicates Removed : 0
Future Orders Removed : 10
Negative Prices Removed : 5
Negative Quantity Removed : 10
Invalid Discounts Removed : 5
Invalid Unit Prices Removed : 5
Invalid Order IDs Removed : 46
Invalid Product IDs Removed : 8

Final Dataset Sizes

Customers : 500
Products : 495
Orders : 490
Order Items : 1426

Cleaned CSV Files Saved Successfully!
=====
RUNNING TEST CASES
=====
Orphaned Order Items : PASSED
Discount Validation : PASSED
Zero Quantity : PASSED
Future Order Date : PASSED
Missing Customer : PASSED

=====
TEST SUMMARY
=====
Passed : 5
Failed : 0
Total : 5

All test cases passed successfully.

```

Flg : Automated Validation Test Results Generated using run_tests.py

Conclusion

The E-Commerce Analytics System was successfully developed and tested using Python, Pandas, SQLite, and SQL. The project demonstrates an end-to-end analytics pipeline including data generation, cleaning, database implementation, SQL analysis, reporting, and automated validation. All modules executed successfully, and all validation test cases passed, confirming the reliability of the implemented solution.